

M003 Week 2 — English Worksheet

Topic: Lines and Grids · **Course:** 3D Coordinates · **Time:** about 45 minutes

This week you place a **line of blocks with a loop**. One number stays the same, the other grows by 1 each time. With lines you can also make a **square outline**.

1 · Predict 🧠

Read each set of steps. Before you imagine it happening, write what you think you will see.

```
set z to 0
repeat 5 times:
  place stone block at (2, z)
  add 1 to z
```

Where does the line start? Where does it end? Which number stays the same?


```
set x to 0
repeat 4 times:
  place gold block at (x, 3)
  add 1 to x
```

Which way does this line grow — along x or along z? How long is it?


```
set z to 0
repeat 3 times:
  place stone block at (0, z)
  place stone block at (4, z)
  add 1 to z
```

Each turn of the loop places two blocks. What do the blocks make — one line or two lines? How far apart are they?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The line should grow along **z**, from (2, 0) to (2, 4). x should stay at 2. Right now the line grows the wrong way.

```
set n to 0
repeat 5 times:
  place stone block at (n, 2)
  add 1 to n
```

Hint: look at which slot n is in — the x slot or the z slot.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The line should be **6 blocks long**, from (0, 0) to (0, 5). Right now it comes out one block short.

```
set z to 0
repeat 5 times:
  place gold block at (0, z)
  add 1 to z
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — This should make a line of 4 stone blocks starting at (3, 0). Right now all the blocks land in the **same spot**.

```
set z to 0
repeat 4 times:
  place stone block at (3, z)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your homework world. Build a **square outline** — like a fence for a garden — using loops to place the lines. When you finish, come back here.

Send a photo or video of your square, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I ...

My loop repeated ... times because ...

The number that stayed the same was ...

The number that grew was ...

One tricky moment was when ...

If I had more time, I would ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while you show your square in the world. Talk through it like you are teaching someone who has never seen it. Try to use these words: **loop**, **line**, **repeat**, **coordinate**, **grow**.

1. Show your square outline, then show the code that built it.
2. Read one loop out loud and say which number grows.
3. Point at the start and the end of one line and say their coordinates.
4. Say in your own words why a loop is better than placing every block by hand.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.